

RATE OF CHANGE
OF ENERGY
INSIDE THE
FLUID ELEMENT

(A)

=

NET FLUX
OF HEAT
INTO THE
FLUID ELEMENT

(B)

+

RATE OF WORK
DONE BY THE
FORCES
PRESENT

BODY
FORCES

(C)

+

PRESSURE
FORCES

(D)

+

SHEAR AND
NORMAL
STRESSES

(E)